CrysDiS

Crystal Diffraction Simulator

(created using ChatGPT Codex)

Start the program

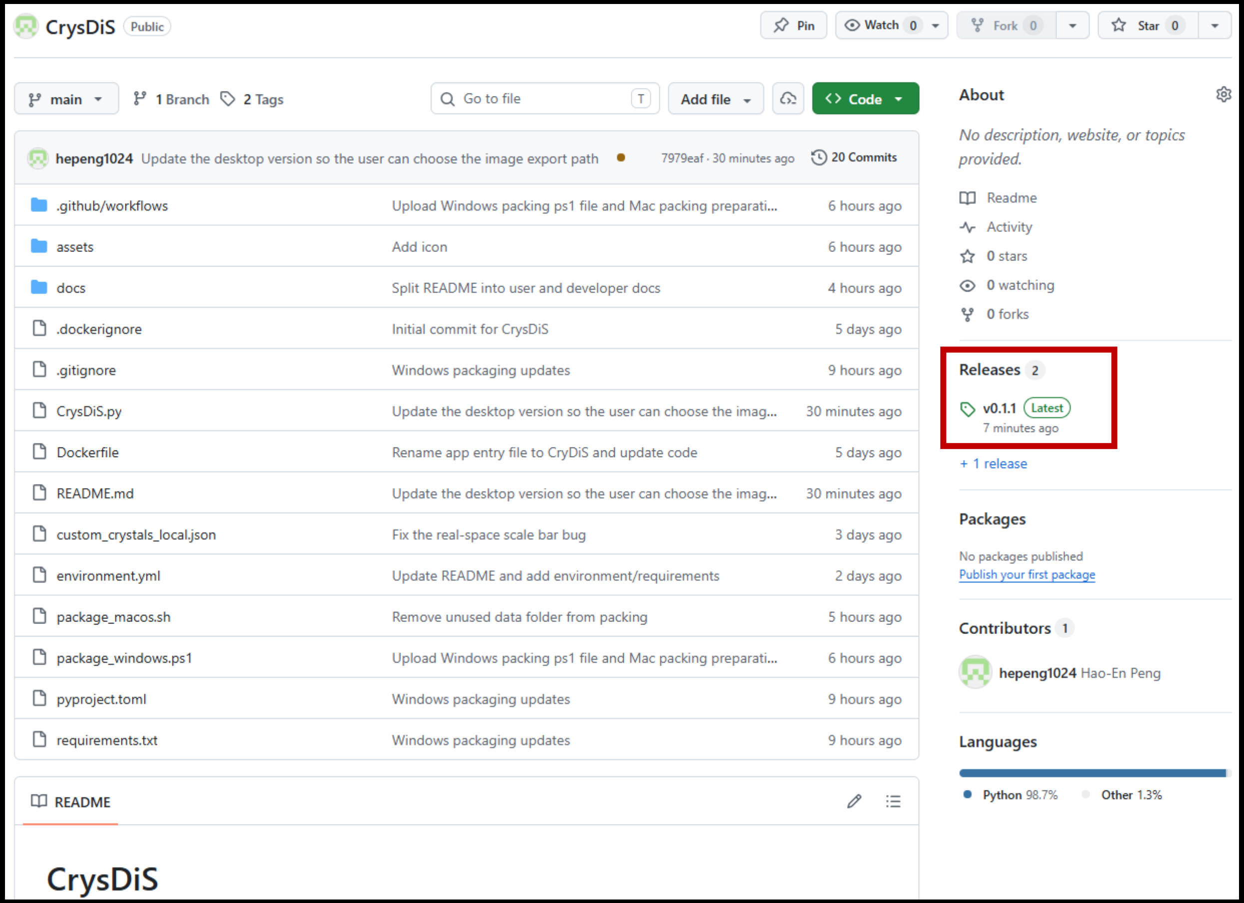
Option 1: use the website version

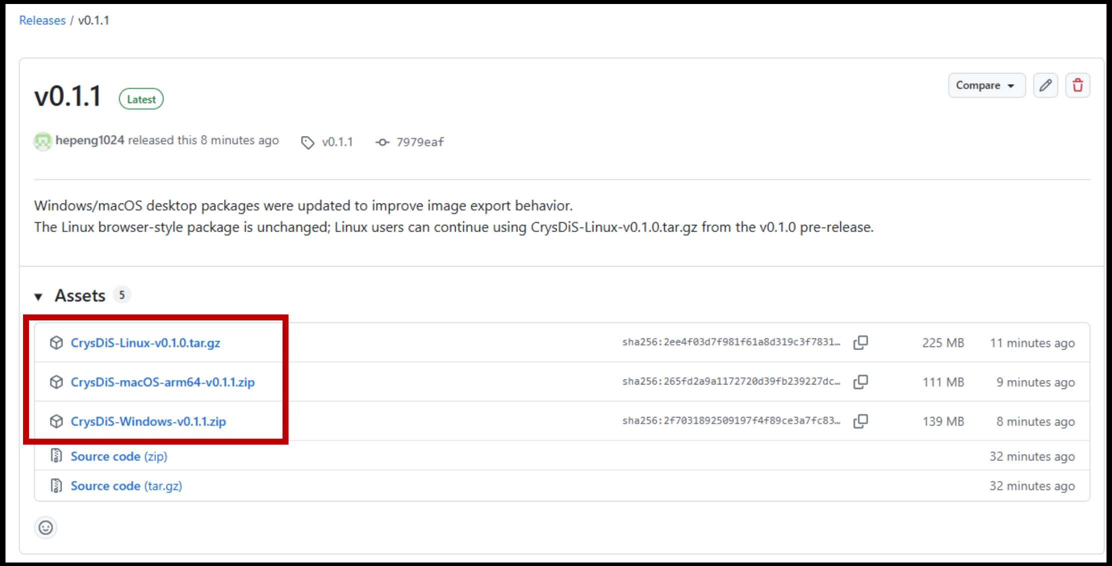
- Website: https://crysdis.onrender.com/
- Since this is a free URL, it may take 2~3 minutes to warm up
- The website version is not stable and may disconnect easily
- If you think this software is useful, running it locally is strongly recommended

Start the program

Option 2: download the program and run it locally

- Go to https://github.com/hepeng1024/CrysDiS → Click "Releases" → Download the version that matches your OS





- Please refer to the README file if you need more information

Start the program

Option 3: Run the source code from GitHub

- First install Git and Anaconda/Miniconda. Then clone the repository:

```
git clone https://github.com/hepeng1024/CrysDiS.git
cd CrysDiS
```

- Create the conda environment:

```
conda env create -f environment.yml
conda activate crysdis
```

- Start CrysDiS:

```
python CrysDiS.py
```

- Then open this address in a browser:

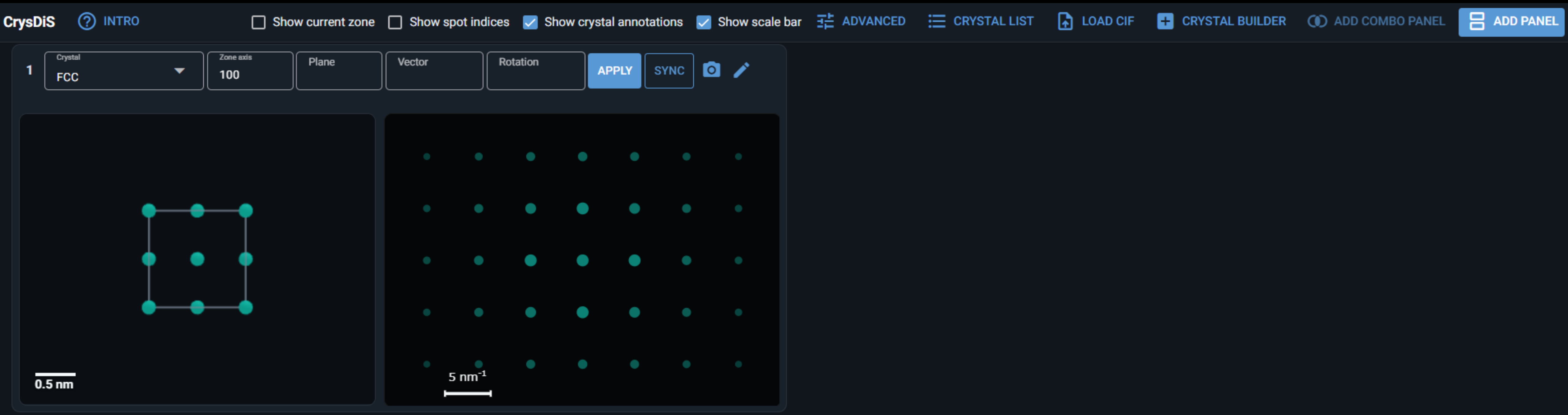
http://127.0.0.1:8080

- If port 8080 is already being used, choose another port:

```
NICEGUI_PORT=8094 python CrysDiS.py
```

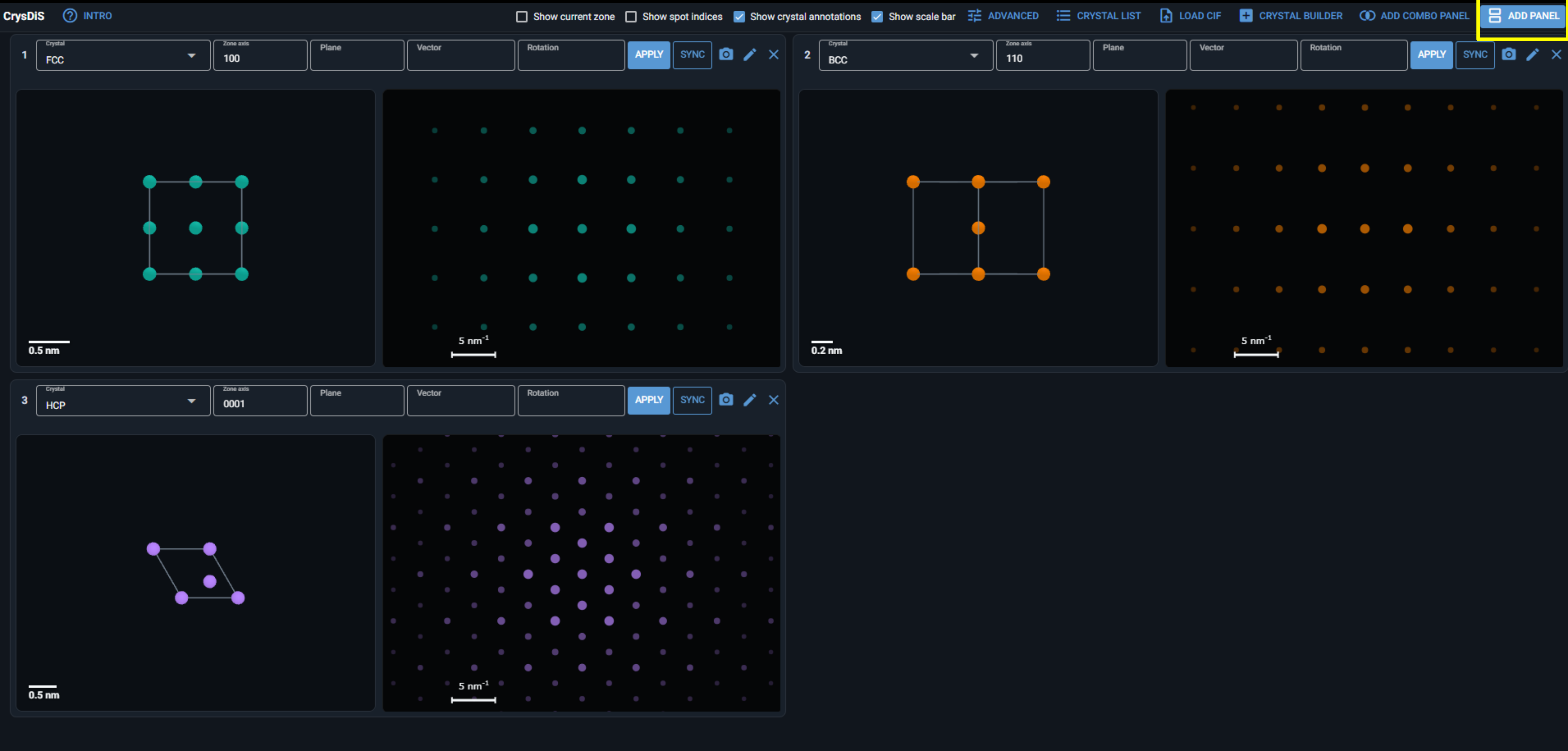
- Please refer to the README file at https://github.com/hepeng1024/CrysDiS for more details

Starting interface



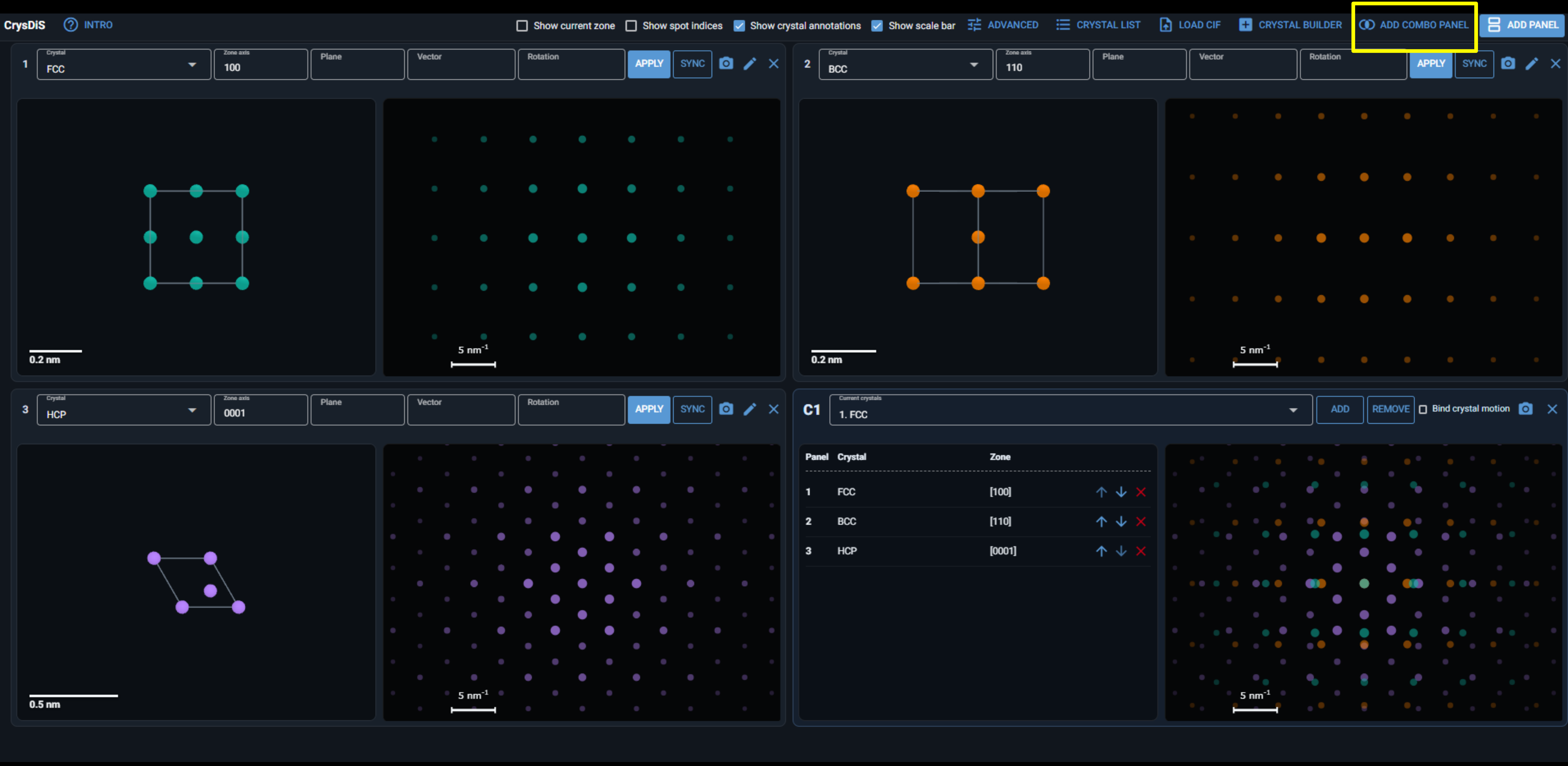
Features of the top toolbar

"ADD PANEL" allows you to have multiple structures at the same time



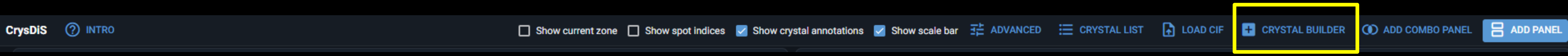
Features of the top toolbar

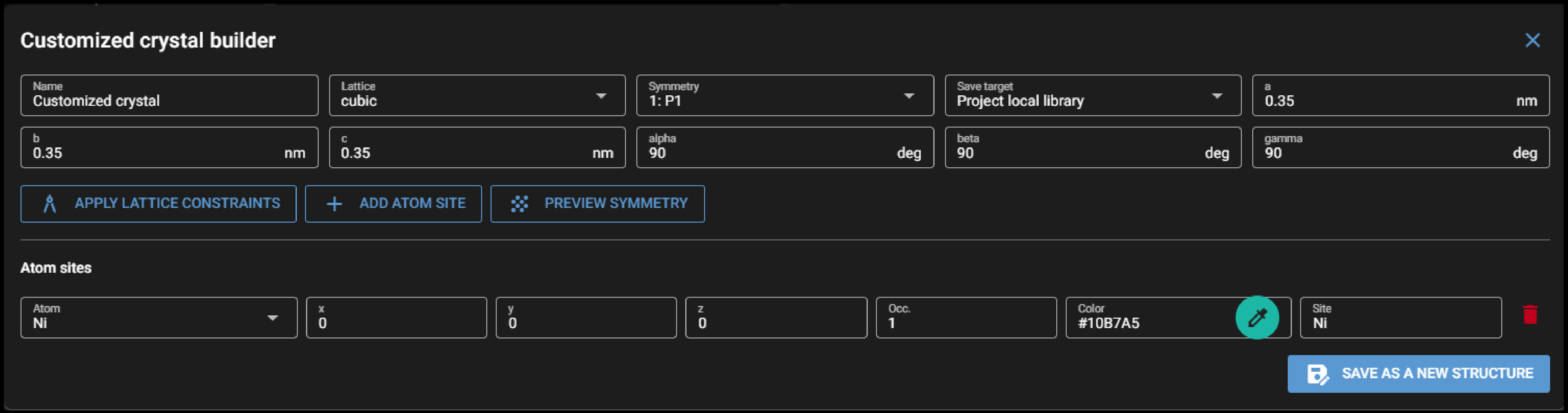
"ADD COMBO PANEL" allows you to overlap diffraction patterns from different panels and compare them



Features of the top toolbar

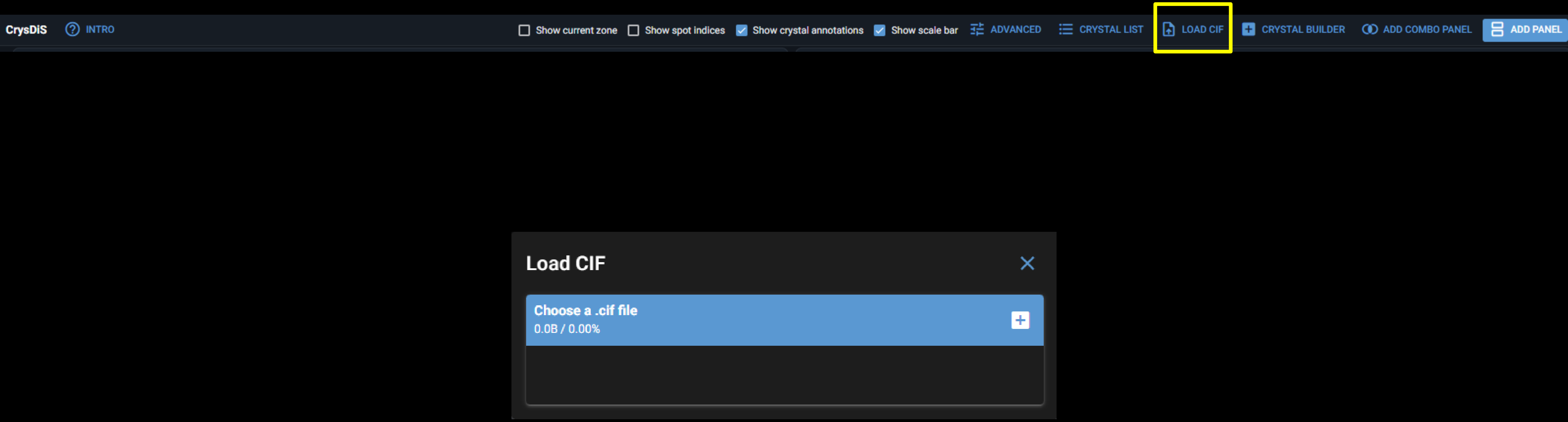
“CRYSTAL BUILDER” lets you build your own crystal





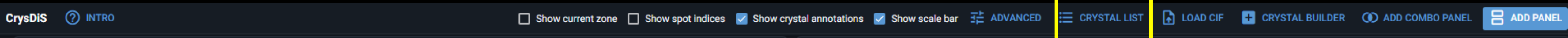
Features of the top toolbar

You can load your own .cif files via “LOAD CIF”

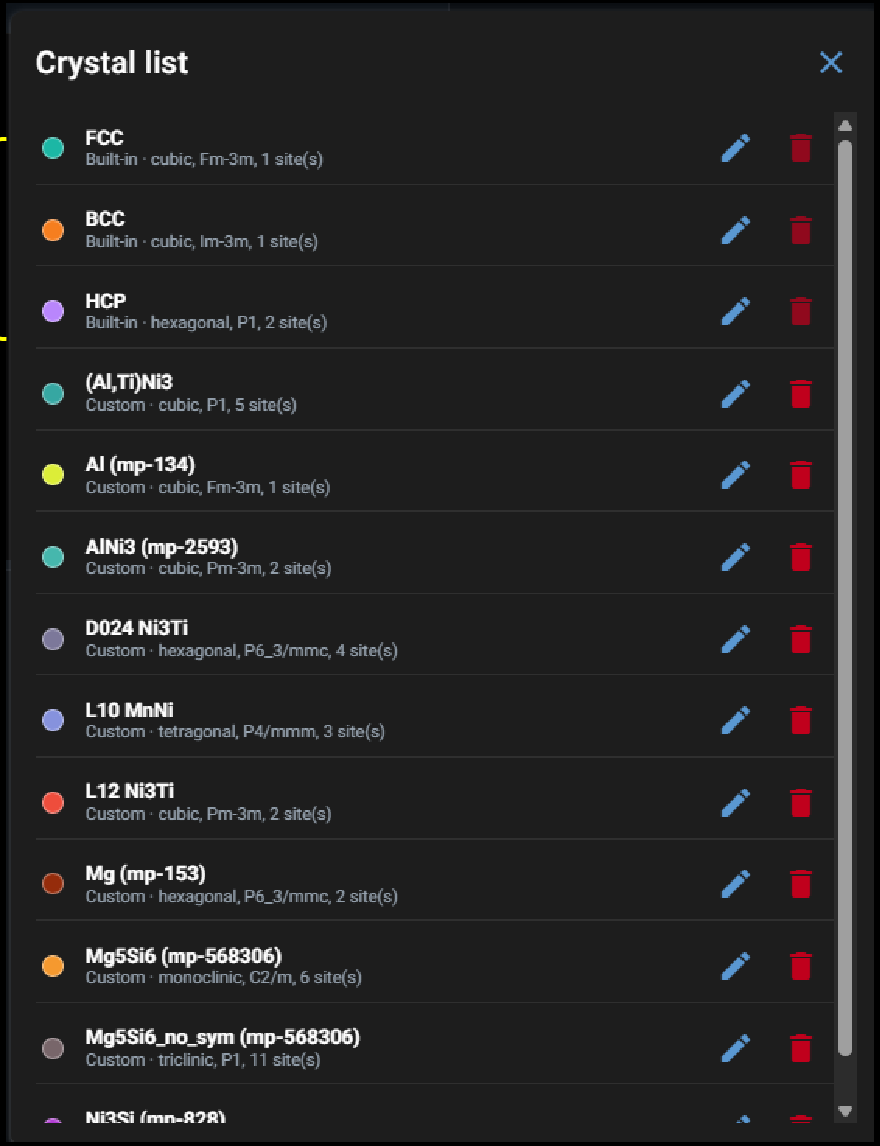


Features of the top toolbar

Manage the crystal structures via “CRYSTAL LIST”

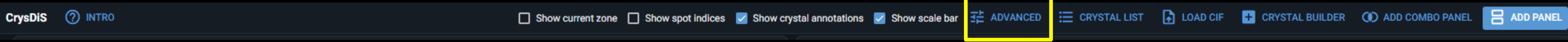


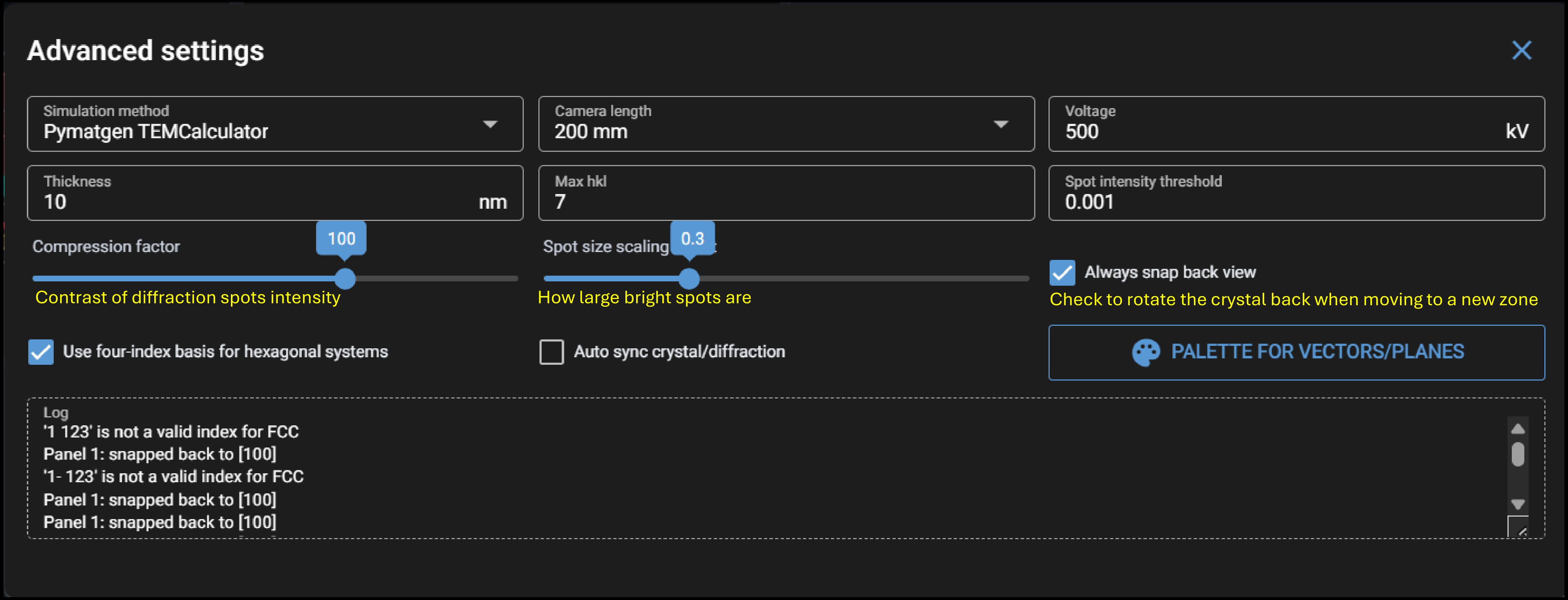
Three built-in structures that cannot be edited



Features of the top toolbar

Advanced settings in “ADVANCED”

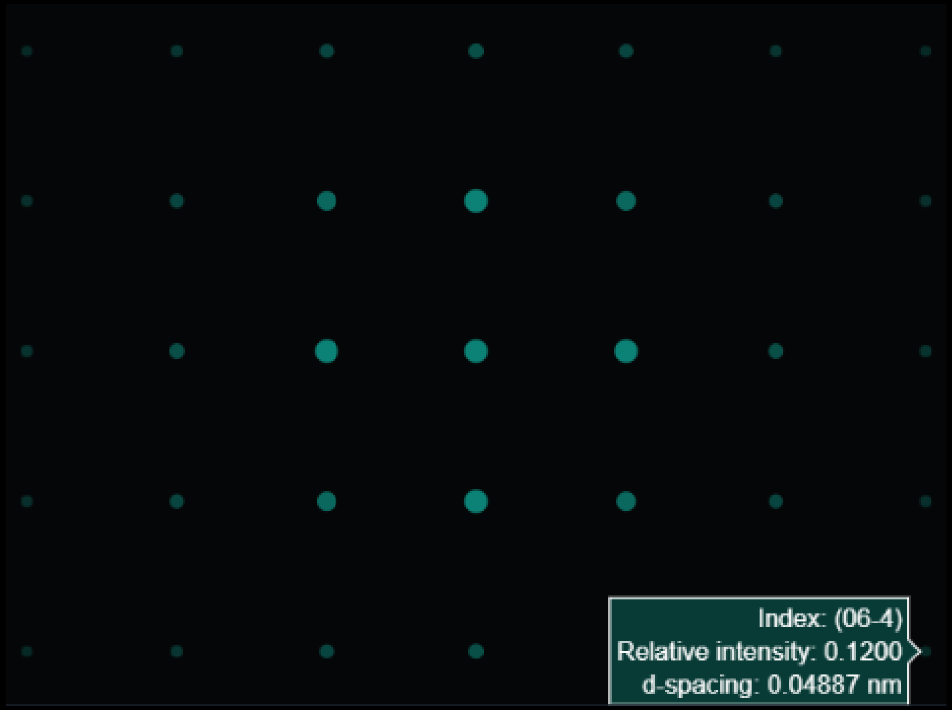




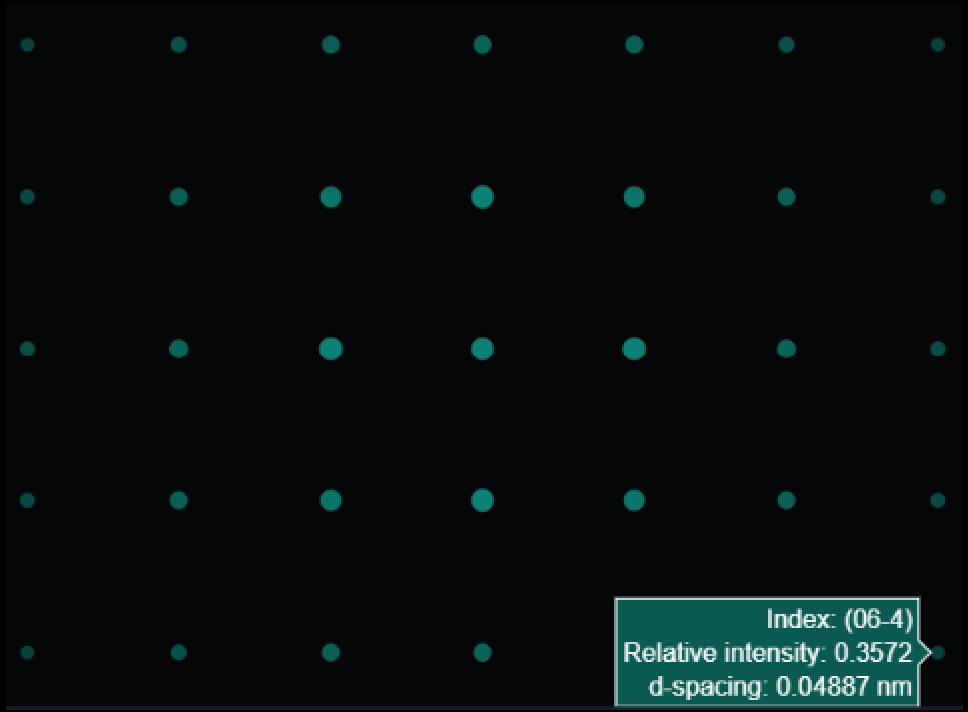
More about advanced settings

- Compression factor

Compression factor = 10



Compression factor = 150



Dim spots are more visible

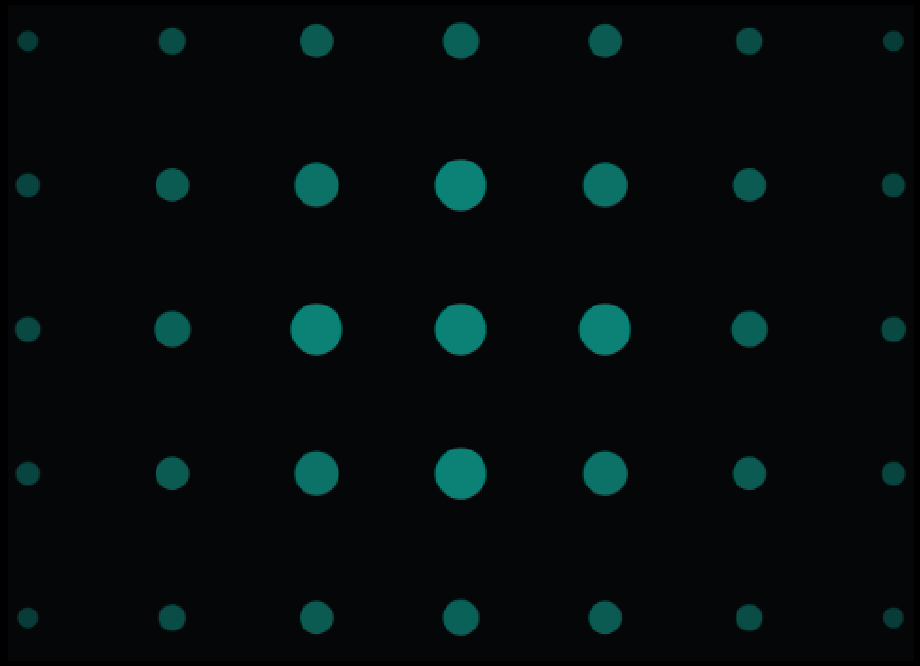
More about advanced settings

- Spot size scaling effect

Scaling factor = 0

All spots are the same size

Scaling factor = 1

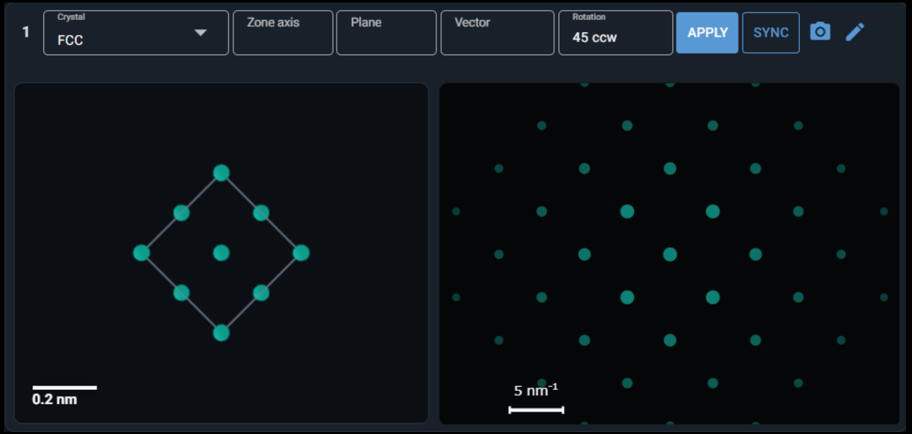


Spot size scales significantly with intensity

More about advanced settings

- Always snap back view

In-plane rotation by 45 deg



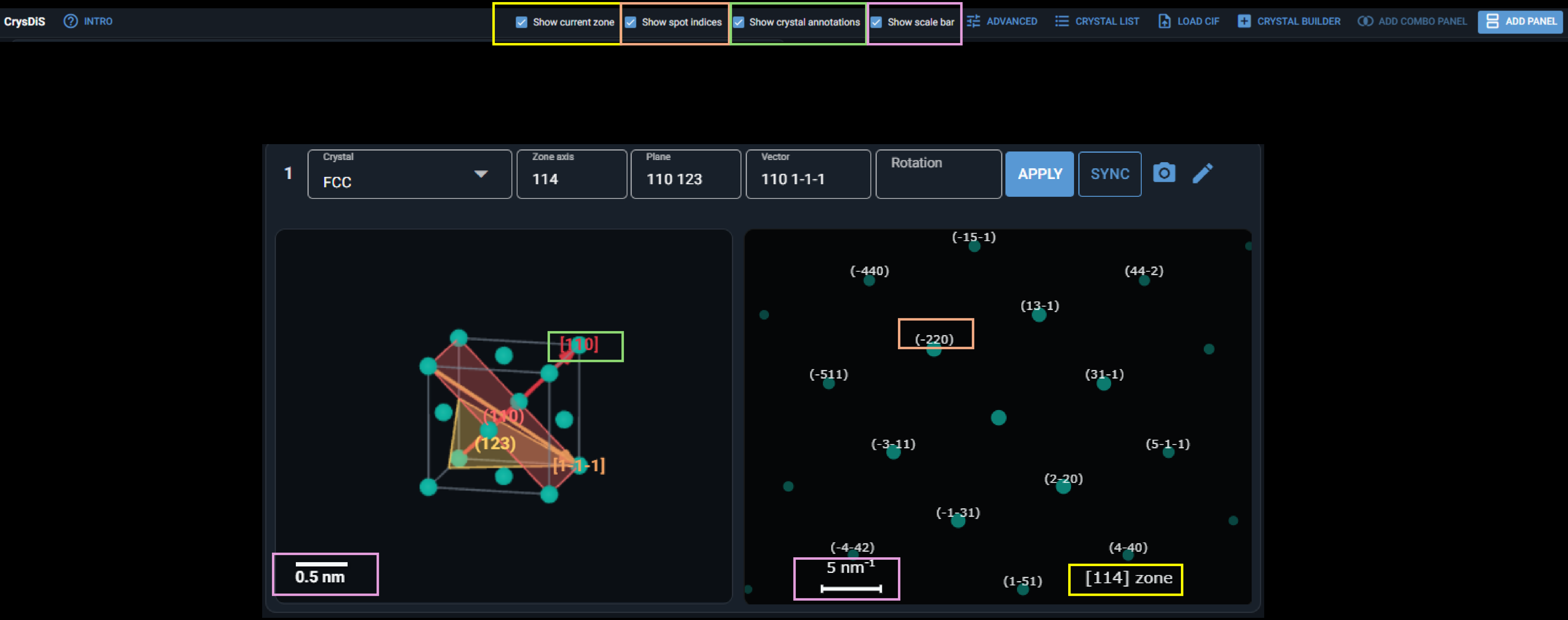
Go to the 110 zone

“Always snap back view” is checked

“Always snap back view” is unchecked

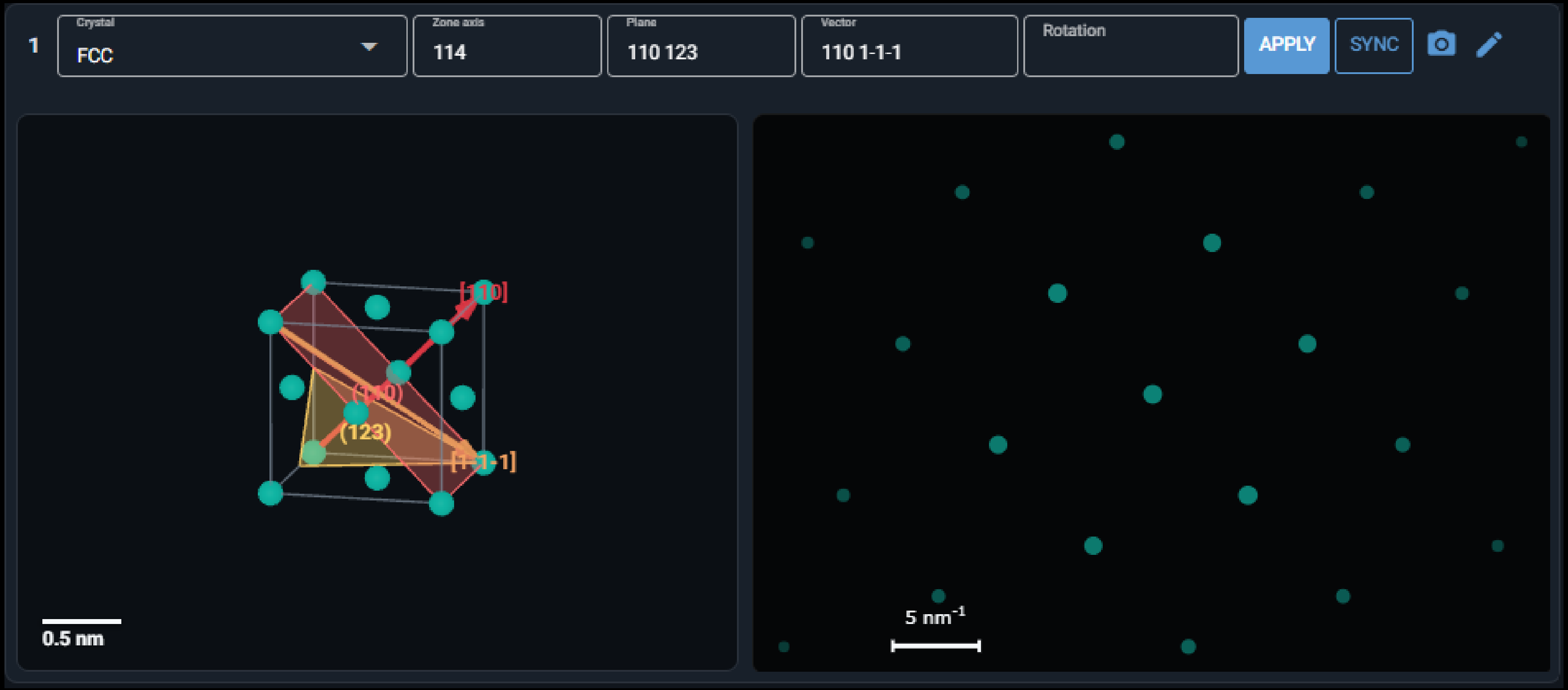
Features of the top toolbar

Some annotation toggles



Features of the panel toolbar

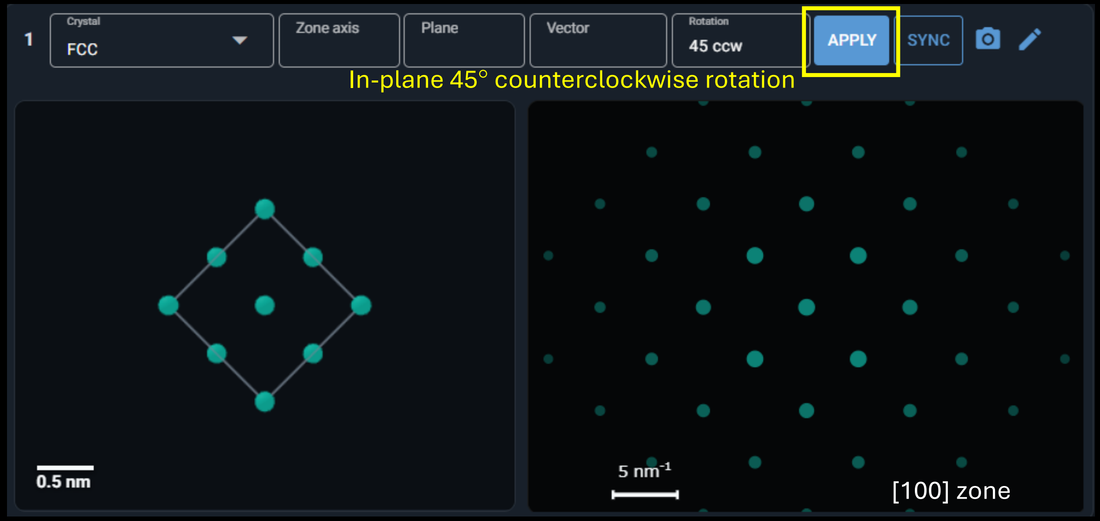
- Choose any crystals in the list or upload/build a new one
- You can go to any zone axis, create vectors/planes
- Reciprocal lattice vectors can be created using *hkl (ex. *110, *0-10)
- For hexagonal systems, 4-index basis inputs are allowed (ex. 2-1-10, *0001)



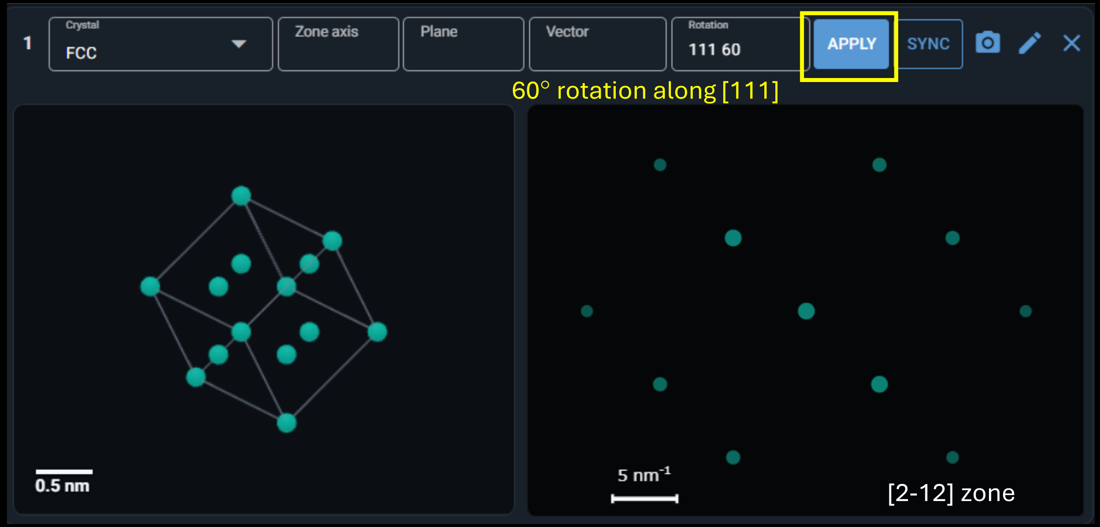
Features of the panel toolbar

- You can also rotate the crystal along a certain axis
- Right-hand rule is applied

In-plane rotation: use “deg cw” or “deg ccw”

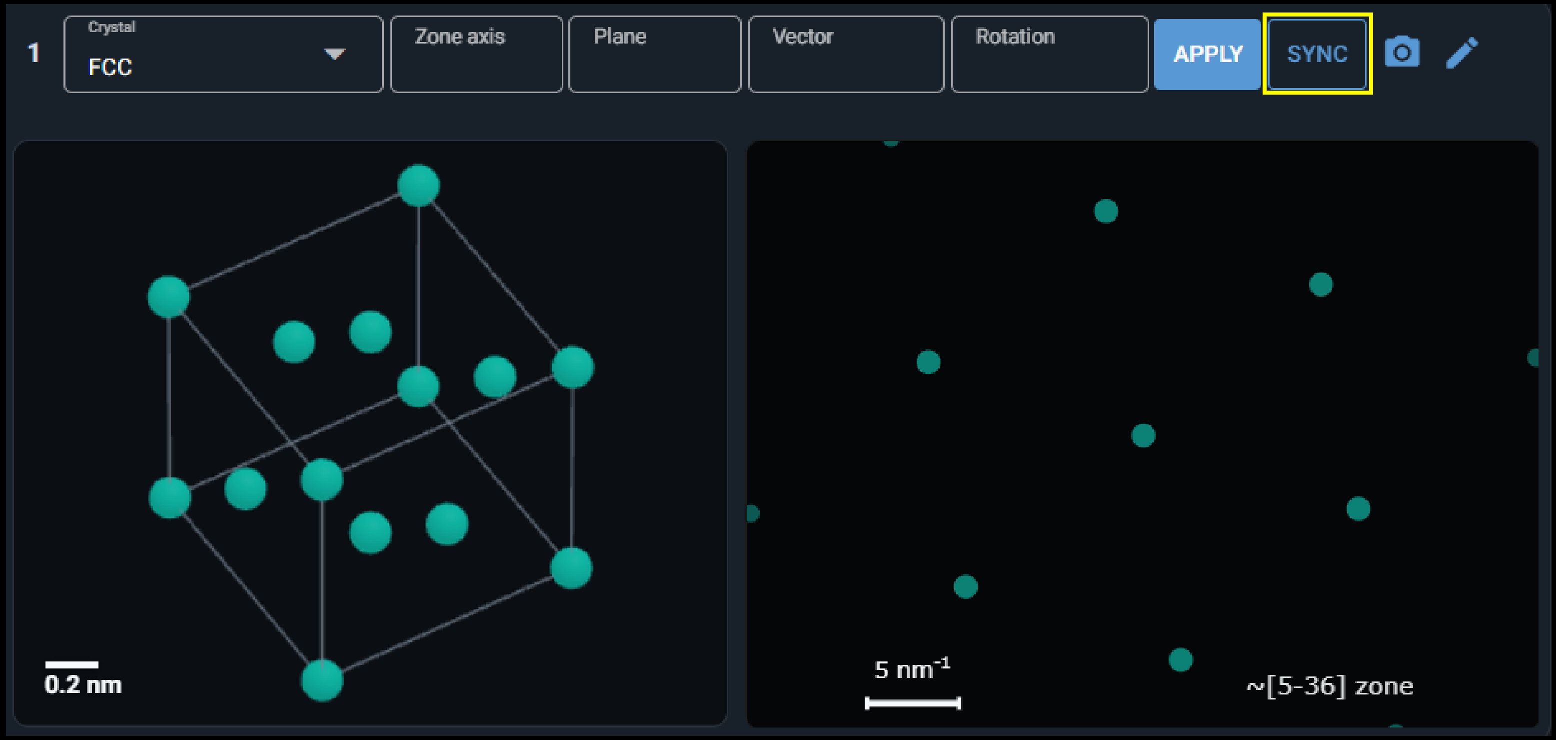


Rotation along a certain axis: use “vector deg”



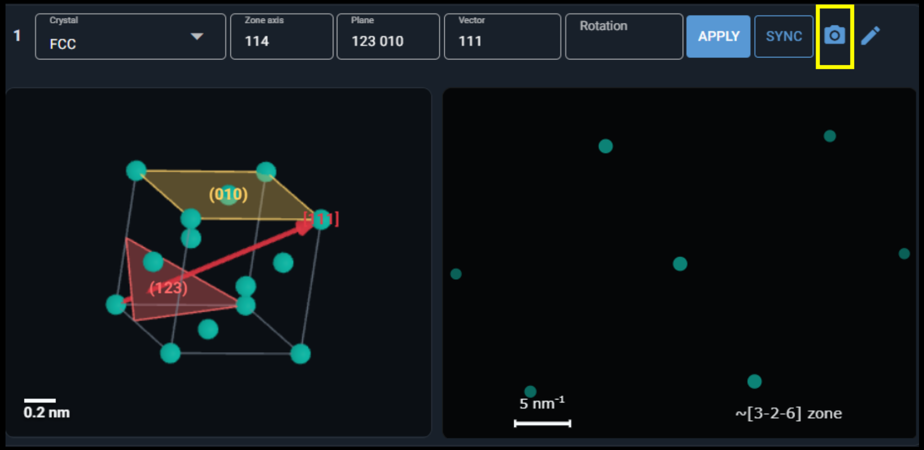
Features of the panel toolbar

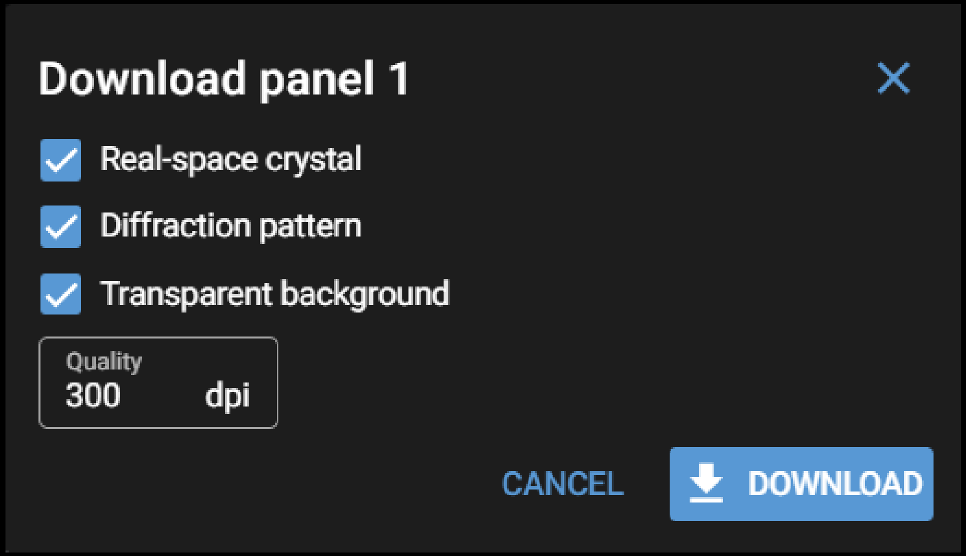
- Synchronize the crystal and diffraction pattern using "SYNC"
- It will detect the closest zone axis
- You can also turn on the "Auto sync crystal/diffraction" in the advanced settings

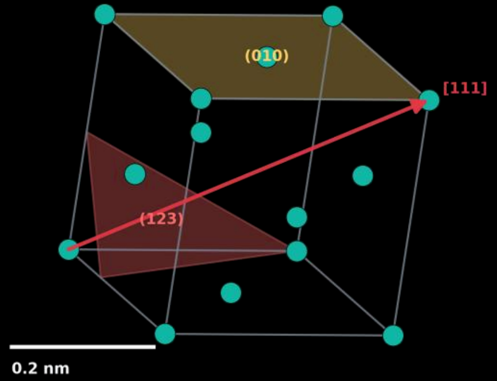


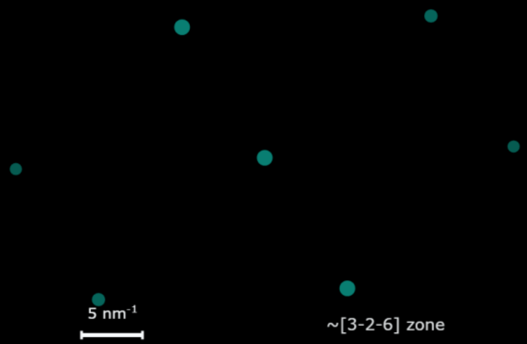
Features of the panel toolbar

- Download the crystal/diffraction pattern at the current viewing angle



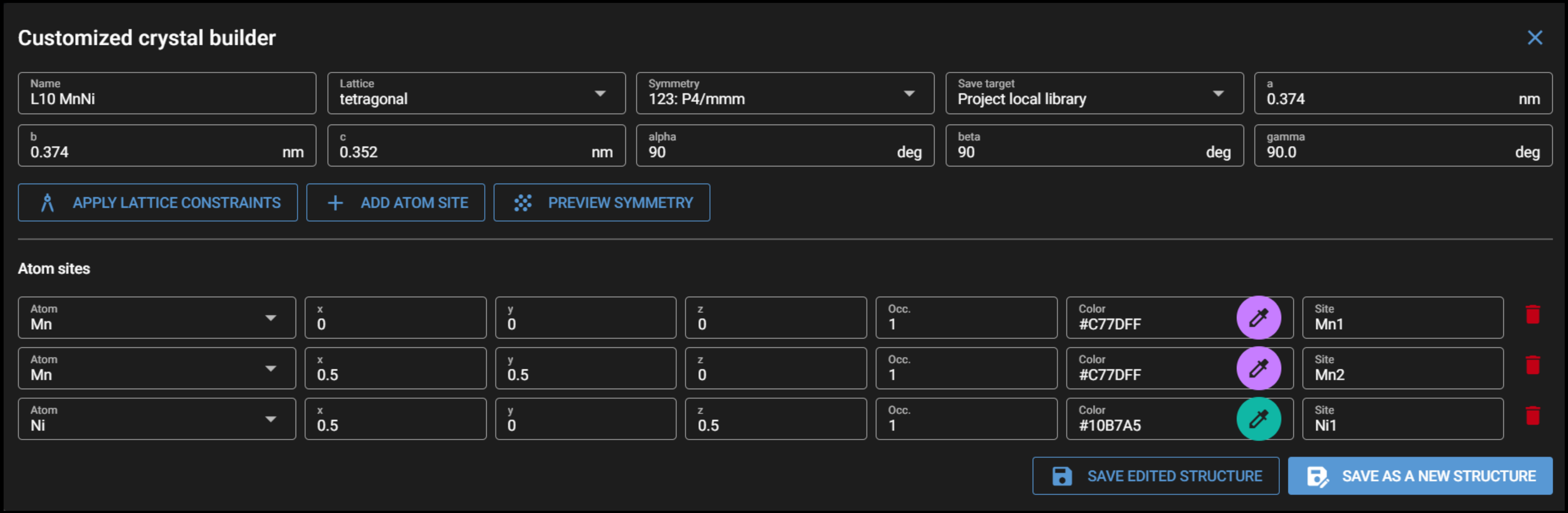






Features of the panel toolbar

- Edit the current structure

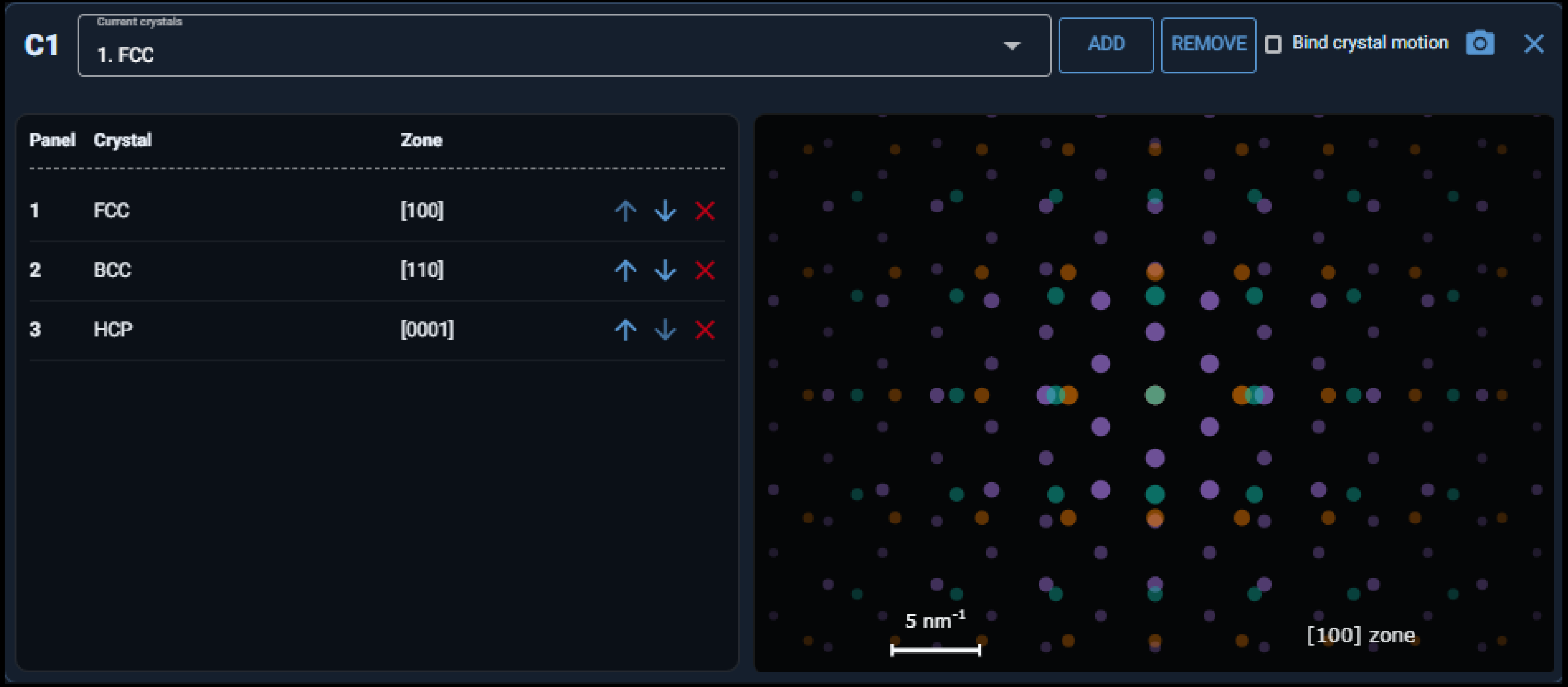


Features of the combo panel

Select crystals present on the page

Add or remove it

The order of the crystal in the list determines the order of the diffraction patterns



Ex. FCC diffraction pattern will be shown on top

The current zone shows the zone axis of the crystal on the top-most layer

Features of the combo panel

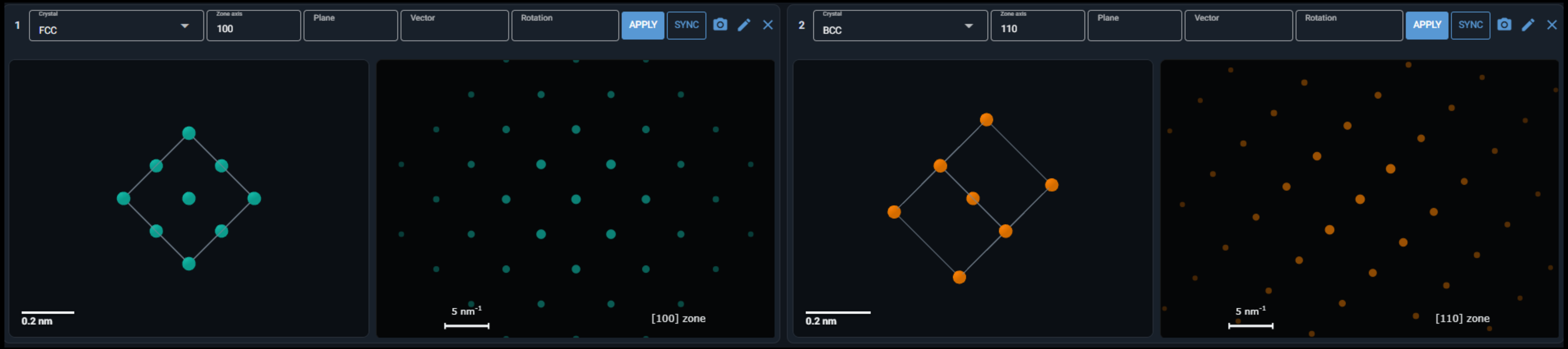
Check “Bind crystal motion” if you want the crystals to synchronize their motions

Features of the combo panel

- Bind crystal motion

Ex. if you rotate the crystal in panel 1 by 45° ccw

the crystal in panel 2 will do the same motion



In this case, it means crystals 1 & 2 have the orientation relationship: $[100]_{crystal1}$ // $[110]_{crystal2}$

Comparison of two simulation methods

Simulation method	Ewald Sphere Kinematic	Pymatgen TEMCalculator
Main purpose	More TEM-like kinematic spot visibility	Fast crystallographic zone-axis pattern
Uses voltage?	Yes, through electron wavelength and Ewald sphere radius	Only for wavelength calculation
Uses thickness?	Yes	No
Includes relrod / excitation error?	Yes	No
Includes Ewald sphere curvature?	Yes	No
Good for	Showing why spots appear/disappear with tilt/thickness	Clean ideal zone-axis indexing/intensity